Shubha Swarnim Singh

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Education

The College of Idaho, Computer Science and Business Administration — August 2022 – May 2026

- Gibson Honors Student, Dean's List, Honor's GPA: 3.77/4.0

Experience

Data Science and Analyst Intern, Albertsons Companies — June 2025 – August 2025

- Built supervised machine learning classification models in Python using Random Forest and XGBoost to predict store profitability across 2,200+ locations, integrating structured datasets, regional features, and sales KPIs.
- Developed ML-driven profitability pipeline analysis with Tableau and Power BI visualizations, supporting $700M+ sales decisions and guiding CAPEX planning in the merchandising department.
- Automated store segmentation and break-even analysis using Python (pandas, NumPy), reducing manual analysis time by 87.5%

IT Intern, College of Idaho — Dec 2022 – May 2023

- Deployed Multi-Factor Authentication (MFA) system across 1,000+ users, achieving 90% adoption rate with secure token validation.
- Managed SQL databases with 1M+ records, writing optimized SQL queries and data integrity scripts to support research and analytics reporting.

Alumni & Events Technician, College of Idaho — Dec 2022 – Present

- Supported alumni fundraising events ($1.1M and $1.3M raised), managing outreach and database-driven targeting.
- Enhanced alumni database, improving engagement and retention by 20%.

Projects

Senate Voting System — **GitHub**

- Engineered a secure full-stack online voting system using Node.js, Express, SQLite, and WebSockets to digitize Senate voting for a 20-member student body, replacing legacy manual processes.
- Built real-time vote visualizations with <50ms latency using WebSockets and D3.js for instant, transparent updates.
- Added role-based access and authentication, enabling secret ballots and cutting groupthink by over 90%.

In-Memory Search Engine — **GitHub**

- Developed a high-performance Python search engine indexing 3,000+ .txt files (5̃00MB) with <10ms latency.
- Implemented data indexing algorithms to reduce search time by 90%, improving scalability and retrieval efficiency.

Investment Calculator — **GitHub**

- Built Python tool leveraging Yahoo Finance API + sentiment analysis to analyze 2,500+ NASDAQ stocks.
- Designed data pipelines for real-time data extraction, processing, and predictive modeling, supporting data-driven investment decisions.

Leadership

- **ASCI Treasurer (2024/25)** & **Senator (2022-2024)**: Managed $100K budget, ensuring effective fund distribution and financial transparency; previously served as a Senator.
- **Consul, Sigma Chi (2025-Present)**: Led the fraternity, overseeing operations and brother development.
- **Resident Assistant (2022-Present)**: Built inclusive community, resolved conflicts, and mentored peers.
- **President, Boxing Club (2023-Present)**: Restarted and grew the club and built a supportive fitness community.

Technical Skills

Programming & Development: Python, JavaScript, SQL, Node.js, Express, Flask, HTML/CSS, Bootstrap, WebSocket
Data Science & Machine Learning: XGBoost, Random Forest, K-means, Genetic Algorithm, SHAP, Scikit-learn
Coursework: Data Structures, Algorithms, Machine Learning, Databases, Programming Languages, Formal Automata